

Rearrangement of a formula



1 Make x the subject of the following equations:

a $y = x + z$

b $y = \frac{x}{-6}$

c $y = 7(2 + x)$

d $\frac{x}{2} + \frac{n}{5} = 2$

e $3y = 2x + z$

f $y = \frac{-x}{7}$

g $y = 5(3 - x)$

h $\frac{x}{5} - \frac{m}{7} = 2$

2 Make m the subject of the following equations:

a $\frac{m}{y} = gh$

b $y = 4mx - 9$

c $x = 5k(n + m)$

d $\frac{2m}{p} = st$

e $b = -2am + 10$

f $x = -7m(p + q)$

g $y = -5mx - 14$

h $x = -7k(l - m)$

3 Make y the subject of the following equations:

a $\frac{x}{y} = h - k$

b $9x^2 + 3y = 12$

c $\frac{3a}{b} = 2j - y$

d $4x^2 - 2y = -16$

4 Make k the subject of the following equations:

a $m = \frac{9kx}{8} - 6$

b $r = \frac{k}{k - 9}$

c $v = \frac{5kw}{7} - 3$

d $s = \frac{2k}{k + 4}$

5 Make R the subject of $V = IR - E$.

6 Make r the subject of $L = \frac{E}{R + r}$.

7 Make p the subject of the equation $M = pq - 5pr^2$.

8 Consider the Pythagorean formula $c^2 = a^2 + b^2$.

a Make a^2 the subject of the formula.

b Make b^2 the subject of the formula.

9 Make r^2 the subject of the following formulas:

a The volume of a cone is given by the formula $V = \frac{1}{3}\pi r^2 h$.

b The area of a circle is given by the formula $A = \pi r^2$.

c

The area of a sector of a circle is given by the formula $A = \frac{\theta \pi r^2}{360}$.

10 Make v^2 the subject of the formula for kinetic energy, $K = \frac{1}{2}mv^2$.

11 The bend allowance for sheet metal is given by the formula $B = 2\pi \left(R + \frac{T}{2} \right) \times \frac{A}{360}$.

Make T , the thickness of the sheet, the subject of the formula.

Applications

12 Temperature conversion from degrees Fahrenheit to degrees Celsius is given by the formula $C = \frac{5}{9}(F - 32)$.

The temperature inside a freezer is 5°F . Find this temperature in degrees Celsius.

13 The perimeter of a rectangle is given by the formula $P = 2(a + b)$.

Find the value of b if $a = 5$ and $P = 26$.

14 The surface area of a rectangular prism is given by the formula $S = 2(ab + bh + ah)$.

Find the value of h if $a = 3$, $b = 3$ and $S = 138$.

15 The surface area of a rectangular prism is given by the formula $S = 2(lw + wh + lh)$, where l , w and h are the dimensions of the prism.

Find the surface area of a rectangular prism with a length of 8 cm, a width of 9 cm and a height of 5 cm.

16 The surface area of a cylinder is given by the formula $S = 2\pi r^2 + 2\pi rh$.

Find the surface area of a cylinder with a diameter of 10 cm and a height of 25 cm.

17 The volume of a sphere is given by the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Find the radius of a sphere with a volume of 124 mm^3 to two decimal places.

18 Newton's second equation of motion is $s = ut + \frac{1}{2}at^2$.

Find the value of a if $s = 2596$, $u = 250$ and $t = 22$.

19 The power of a circuit is given by the formula $P = I^2 R$.

Find the value of I , correct to two decimal places, when $P = 40$ and $R = 560$.



- 20** The resistance of a parallel circuit is given by the formula $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$.

Find the value of b , correct to two decimal places, if $a = 8$ and $c = 23$.

- 21** The reactance of a capacitor is given by the formula: $X = \frac{1}{2\pi fC}$.

Find the value of f , correct to one decimal place, if $X = 44.1$ and $C = 190 \times 10^{-9}$.

- 22** The gravitational force between two masses is given by $F = \frac{GMm}{d^2}$.

Find the value of M , correct to two decimal places, if $F = 15$, $G = 6.67 \times 10^{-11}$, $m = 1$ and $d = 5 \times 10^{-6}$.

- 23** Amelia bought a television series online. The television series was 4.8 gigabytes large and took 4 hours to download.

a Find the size of the television series in gigabits. Use the formula $b = 8B$, where b is the number of gigabits and B is the number of gigabytes.

b Hence, find the average number of gigabits downloaded each hour.

- 24** The life expectancy, y , of a person born in Japan is approximated by the equation $y = 0.27x + 72$, where x is the number of years since 1970.

In what year will the life expectancy in Japan reach 95 years?

- 25** The amount of oil, y , imported to the United States from Canada in millions of barrels per day can be approximated by the equation $y = 0.105x + 1.34$, where x is the number of years since 2000.

In what year will the approximate number of barrels imported from Canada be 5.5 million per day?